

Nguyen Trong Hieu

Email: hieutnguyenengineer@gmail.com | Google Scholar: <https://shorturl.at/g0OhF>

EDUCATION

Korea Advanced Institute of Science and Technology (KAIST)

Daejeon, South Korea

Master of Science Mechanical Engineering

Aug 2023 – Aug 2025

- GPA: overall 3.57/4.3 (92.7/100); major-only 3.67/4.3
- Thesis: “Learning Loco-manipulation Control for Humanoid Robots via Action-pretrained Reinforcement Learning”
- Advisor: Prof. Hae-Won Park

Hanoi University of Science and Technology (HUST)

Hanoi, Vietnam

Bachelor of Science in Control Engineering and Automation

Sep 2018 – Aug 2022

- GPA: overall 3.75/4.00 (9.38/10) – Rank 2/32 students in the *Talented's Bachelor Program*
- Thesis: “A Disturbance-observer based Safety-critical Control for Uncertain Affine Nonlinear Systems”
- Advisor: Prof. Nam Hoai Nguyen

RESEARCH EXPERIENCE

Learning Loco-manipulation Control for Humanoid Robots via Action-pretrained Reinforcement Learning

*Conducted at the **Dynamic Robot Control and Design Laboratory**, KAIST*

- Developed a learning-based control framework for humanoid robots to perform a loco-manipulation task. The learning environment is built upon the massively parallel framework introduced in IsaacGymEnvs.
- Trained policy utilizing the *Action Pretrained Transformer based Reinforcement Learning* framework to produce more efficient motion than naïve Reinforcement Learning.
- Developed a Trajectory Optimization framework in MATLAB to generate action-state trajectories dataset for the pretraining step.

A Disturbance-observer based Safety-critical Control for Uncertain Affine Nonlinear Systems

*Conducted at the **Multi-agent System Control Laboratory**, HUST*

- Proposed a robust safety-critical control framework combining the *optimal-decay Control Barrier Function - Control Lyapunov Function based Quadratic Programs* and a novel disturbance observer. The proposed method is computationally efficient and explicitly accounts for input saturation during the controller design.
- Provided theoretical analyses to establish both *input-to-state stability* and *input-to-state safety* of the closed-loop system.
- Validated the approach through MATLAB simulations, implementing the *L1 Adaptive Control Barrier Functions* framework as a benchmark. The proposed method outperformed the L1-CBF approach in safety enforcement and tracking performance, while requiring roughly half the computation time.

WORK EXPERIENCE

WIM Inc.

Daegu, South Korea

Robotics and AI Engineer

Aug 2025 – Present

- Implemented and improved the accuracy of the low-level controller for 6-DOF manipulators, making PID tuning easier and more intuitive.
- Conducted research on camera calibration techniques leveraging machine learning for industrial robotic applications, aiming to improve the accuracy and automation of the calibration pipeline.
- *Key skills:* Python, C++, ROS2, Manipulator planning and control, Camera calibration.

Viettel Aerospace Institute

Hanoi, Vietnam

Modeling and Simulation Engineer

Oct 2022 – Aug 2023

- Evaluated kinematic characteristics (for example, achieved range, velocity, maneuverability, time constants) of the unmanned air vehicles (UAVSs) based on their aerodynamics and configurations.
- Designed and developed model-based control algorithms for UAVs by linearizing the UAV model around operating points and designing corresponding linear controllers.
- *Key skills:* MATLAB, Simulink, UAV simulation, system modeling, linear control design.

PUBLICATIONS

H. T. Nguyen, D. Q. Nguyen, and N. H. Nguyen, “A disturbance–observer based safety-critical control for uncertain affine nonlinear systems,” in *International Journal of Control, Automation and Systems (IJCAS)*, vol. 23, pp. 1457–1468, 2025 [[pdf](#)] [[code](#)]

T. T. Bui, T. T. Cao, **T. H. Nguyen**, D. H. Le, H. H. Dao, and P. N. Dao, “Disturbance-observer based reinforcement learning for overhead crane systems,” in *2022 11th International Conference on Control, Automation and Information Sciences (ICCAIS)*, IEEE, 2022, pp. 19–23 [[pdf](#)]

HONORS AND AWARDS

KAIST Scholarship (Full tuition and fees) for the Master Course 2023
Given by KAIST

Best undergraduate graduation thesis award 2022
Awarded by School of Electrical & Electronic Engineering, HUST

Two First Prizes at 23rd National Undergraduate Student Physics Olympiad 2022
Awarded by Vietnam Physical Society

- In the solving problem section and the multiple-choice section.

Outstanding academic scholarships 2021 – 2022
Given by HUST

- Awarded for top students having outstanding academic achievements.

TECHNICAL SKILLS

Domain: Reinforcement Learning, Control Theory, Optimization-based Control, Robot Modeling and Simulation

Programming: Python, MATLAB, C/C++

Physics Engines or Robot Simulation Platforms: IsaacGym, Raisim, Gazebo

Tools: Git, Docker, Simulink, ROS2, Torch, CasADi

Languages: English (IELTS 7.5 — Listening: 6.5, Reading: 8.5, Writing: 7.0, Speaking: 7.0)

EXTRACURRICULAR ACTIVITIES

Participate in the 9th Vietnam Summer School of Science Aug 2022
Sponsored and organized by the Rencontres du Vietnam Foundation and ICISE

- A summer school for young Vietnamese from diverse fields passionate about scientific research.
- Among 146 candidates selected from more than 700 applicants nationwide.

Learning Supporting Club 2019 – 2020
Managed by the Student Union of HUST

- Lecturer for Calculus I and Teaching Assistant (Tutor) for Calculus II.

REFERENCES

Assoc. Prof. Hae-Won Park

Director of Humanoid Robot Research Center [[website](#)], Department of Mechanical Engineering, KAIST

Email: haewonpark@kaist.ac.kr

Assoc. Prof. Nam Hoai Nguyen

Lead of Automatic Control Theory Group, Head of MASC Lab [[website](#)], Department of Automation, School of Electrical & Electronic Engineering, HUST

Email: nam.nguyenhoai@hust.edu.vn